Technical reference manual
1.0.0

EtherBench

Calibration values

The calibration optionnal page target application where the analog input has been adapted for any reason, and some corrections must be applied. As a reminder, the internal ADC of the ST MCU is used, therefore, for precise analog IO, a specialized tool shall be used.

Page structure

Offset	Size (Bytes)	Description
`0x00`	**2**	Pages marker, constant to 0xEBB4
`0x02`	**2**	Page CRC
`0x04`	**4**	ADC 0 Offset (float)
`0x08`	**4**	ADC 1 Offset (float)
`0x0C`	**4**	ADC Gain (float)
`0x10`	**4**	DAC 0 Offset (float)
`0x14`	**4**	DAC 1 Offset (float)
`0x18`	**4**	DAC Gain (float)
`0x1C`	**2**	Calibration year
`0x1E`	**1**	Calibration month
`0x1F`	**1**	Calibration day

Yaml arguments

When using the automated page builder, available on the EtherBenchProbe repo, all of this page shall be placed under the calibration tag. Then, these keys are searched:

Key	Format	Default value
adc0_offset	Any SI value (12mV, 1V...)	`0V`

adc1_offset	Any SI value (12mV, 1V...)	`0V`
adc_gain	Float	`1.0`
dac0_offset	Any SI value (12mV, 1V...)	`0V`
dac1_offset	Any SI value (12mV, 1V...)	`0V`
dac_gain	Float	`1.0`
year	Integer (16 bits)	`2026`
month	Integer (8 bits)	`1`
day	Integer (8 bits)	`1`

The present of any key will trigger the build of the page. Missing keys are going to be filled with the default value.